Voltage-Current Behaviour of Damaged High Temperature Superconducting (HTS) Tapes

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Abstract: The development of High Temperature Superconducting (HTS) cables is vital for high-power transfer applications at cheap liquid nitrogen temperatures when working with Rare Earth Barium Copper Oxide or (RE)BCO HTS tapes. The maximum current that a singular (RE)BCO HTS tape can transfer is too low to be independently useful in a power engineering setting. HTS tapes are more susceptible to damage than traditional copper and they require cooling at cryogenic temperatures in order to superconduct. Connecting HTS cables or equipment to the grid is attractive because such technology can offer extremely low electrical resistance and low voltage when operating at high power. Building an HTS cable with HTS tapes requires an understanding of the tape behaviour when subjected to damaging or extreme conditions. This understanding ensures that such conditions are avoided and that the tapes are sufficiently protected. We have exposed HTS tapes to unfavourable conditions, such as thermal quenching, atmospheric oxidation, bending, and extreme currents. The results have assisted in developing a guiding knowledge base of possible HTS tape behaviour that may occur during the development of an HTS cable. The hope being that this knowledge may add value to prospective HTS cable development processes.

Index terms – HTS, Superconducting, Tape

1 INTRODUCTION

The practical intention to develop and implement a superconductor cable for high power transfer began in 1967, when R. L. Garwin and J. Matisoo proposed a 100 GW (100 kV, 500 kA) DC superconducting cable of 1000 km in length [1], [2]. This proposal, coupled with the discovery and production of various High Temperature Superconducting (HTS) products [3], paved the way for present day and prospective HTS cable initiatives. One such initiative is the use of HTS tapes to construct a multi-tape HTS cable with very-high current capabilities [3]. Such a cable introduces benefits over traditional copper cables especially when considering urban underground cable applications. Superconducting cables present current density and environmental impact advantages. A superconductor cable can transmit much higher power for the same volume of a traditional copper cable. Additionally, the low voltage transmission removes the need to consider the negative environmental impacts of high-voltage transmission.

In 2015 a 500 m, 3.25 kA, 80 kV DC YBCO superconducting power cable was installed on Jeju Island, South Korea [4]. It was the first superconductor power cable to be energised in South Korea's power system and the longest implemented superconductor cable of its time. It was also the first to provide intra-power-system utility as a substation high-power transfer cable. Since then, several DC superconductor cables have been implemented around the world including a 2.5 km cable installed in St. Petersburg, Russia in 2016. It is the longest DC superconductor cable currently in commercial operation [4].

The prospects for applications of HTS tapes in replacing traditional copper within the South African power grid may be numerous and perhaps bound by both practical and financial feasibility. Due to the superconducting tapes' high current and no resistance nominal operation, the most attractive applications would be in high-power transfer and high-current surge applications. Prospective designs around current limiters and power grid reactors could introduce methods for reducing nominal operation losses and improve power grid protection. There may even be relevant superconductor cable applications in high density areas like Sandton, Johannesburg where there is substantial load growth and no new power corridors.

2 BACKGROUND

Measurements performed on HTS tapes often present challenges as the measurement apparatus systems require high electrical sensitivity, noise reduction and resolution [5]. The HTS tape can exhibit unintuitive electrical response variations. Its physical fragility lends it towards small and often overlooked experimental environment variables. Often these variables are unavoidable, as with variations in sample-electrode contact resistance which might be small, but in a near-zero-resistance superconducting environment, play a signification role in current injection and distribution. The tape can also respond in an unexpected manner due to changes in the experimental environment and sometimes unintended damage caused by this environment. Stress-strain damage to an HTS tape can cause the superconducting material to crack which causes degradation of the critical current [6]. HTS tapes have a temperature limit when soldering electrode contacts. If this limit is exceeded, the tape may become damaged which compromises its nominal functionality. In a superconductor research climate which is geared towards the discovery of potentially disruptive technologies and the persistent improvement of existing technologies, it is vital that researchers understand the possible behaviour of superconducting tapes exposed to damaging or extreme conditions. The objective is to ensure that HTS tape research accurately represents the functionality of the HTS tapes and that any functional system developed is protectively contained and monitored to maintain its efficacy. This motivates a need to practically investigate the HTS-tape responses under varying and possibly undesirable conditions prior to practical development of a complete HTS multi-tape cable.

A typical single HTS tape, sourced from SuperPower Inc, SCS4050 (RE)BCO is rated to carry a minimum of 100 A at T = 77 K (boiling point of liquid nitrogen) [7]. A multi-tape cable would need to be constructed for functionality at currents in excess of 500 A, as the tape fabrication technology is incapable of producing a single tape with such a high current carrying rating. A challenge related to multi-tape cables is achieving evenly distributed tape current sharing, as it is difficult to produce tape/current-injection contacts with the exact same contact resistance. The resulting unbalanced current distribution implies a reduction in the cable's critical current.

HTS tapes can exhibit damaged behaviour dependent on environmental exposure, over-heating and/or bend damage. When considering a cable, if the damage is significant, it is important to be able to accurately identify which tape is damaged and where along the length of the tape the damage occurred. An HTS tape immersed in liquid nitrogen that carries a current much higher than its critical current, or that is exposed to a higher than critical temperature condition, more likely will undergo a thermal quench. The quench could result in a damaged or completely destroyed tape. In order to determine the limits and behaviour of HTS tapes under varying conditions, it is necessary to inject current levels in excess of the tape's critical current. Knowledge of how an HTS tape behaves near quenching current is important in order to mitigate these events.

There are undesirable conditions when using a liquid nitrogen cooled bath. These may include bubbling, ice pollution, frosting on or near the HTS tapes and the risk of the level of liquid nitrogen dropping too low. The liquid nitrogen cooling bath method, however, presents a benefit over cryogenic chambers whereby runaway heat and localised heat energy is better absorbed by the 77 K liquid nitrogen medium [5]. This allows for experimentation well above the critical current before a quench occurs.

2.1. The Switching Effect

The *switching effect* can occur due to an alternative normal conducting path that is parallel to the superconductor material [5]. The true curve of an HTS tape is the nominal exponential voltage-current curve of an undamaged tape measured between two voltage taps. Damaged HTS tapes can present a sudden resistance within the superconductor material at currents exceeding the critical current. When this resistance occurs, it forces the current out of the superconductor material and into the normal conducting path (copper stabiliser of the HTS tape). If this happens within a differential voltage tap, the voltage drop will be measured and the voltage-current curve represents that of the tape's true curve. If it occurs outside of the voltage tap, the voltage-current curve will be shifted to a higher current with respect to the true curve [5].

3 APPARATUS EXPERIMENTAL SETUP

The HTS tape apparatus was constructed to allow for easy tape distance adjustments. It was interfaced to a precise nanovoltmeter and a high-accuracy DC power supply specifically designed and manufactured for superconductor experimentation.

3.1. HTS Tape Experiment Apparatus

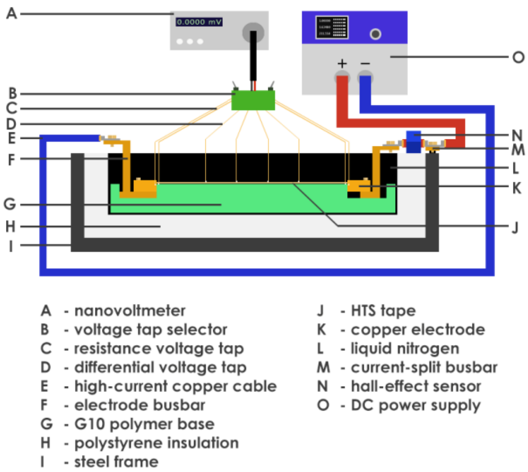

Fig. 1: HTS Tape Experiment Schematic.

The experimental apparatus, shown in Figure 1, consisted of a double stacked Oxford Instruments iPS-M DC power supply (O) rated up to 125 A/10 V total maximum output. This supplied the power via high-current-rated copper cables (E) and electrode busbars (F) to the copper electrodes (K). The HTS tape (J) is securely fixed to the copper electrodes through a specialised process to ensure good electrical contact to reduce input resistance. The input resistance is measured using the resistance voltage tap leads (C).

These leads and the differential voltage tap leads (D) are fed into a custom made voltage tap selector (B) which allowed for the selection of two of any of the leads to be outputted as differential voltage inputs into a Keithley 2182A nanovoltmeter (A). The G10 polymer base (G) was custom designed and machined from a 50 x 12 x 2.5 cm piece of G10. The liquid nitrogen (L) is poured into a stainless steel container surrounded by polystyrene thermal insulation (H) which is supported by a steel frame (I).

3.2. Electrode-HTS-tape Contacts

The HTS tapes were attached to electrodes using a specific soldering method. This method consists of using a heating block to heat up the copper electrodes to a sufficient temperature to melt the solder (~183 ̊ C). The solder amount used is enough to hold the tape but not in excess. The tape is then lightly placed and pressed against the electrode at two points. It is inspected and, whilst still holding the HTS sample in a pressed position, water is sprayed on the HTS tape to ensure it is cooled and retains its position. This results in a good electrical contact with the electrode and minimises the transfer distance. See Figure 2.

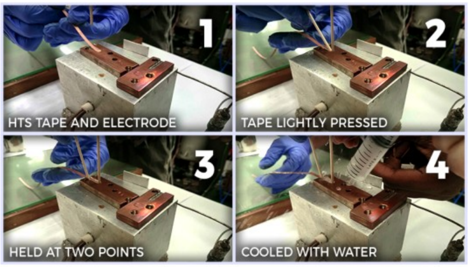

Fig. 2: Step-by-step soldering method process for soldering HTS tapes to copper electrodes.

An extra voltage lead is soldered to the contact electrodes during the contact method process. This voltage lead is used to measure the contact resistance. It can also be used to monitor the potential difference between the contact electrode and any voltage tap on the superconductor [5].

3.3. Voltage Taps

Voltage taps consisted of thin 0.1 mm laminated copper wire which avoided conducting heat into the experimental apparatus. The HTS tapes were populated with six voltage taps: two for the input and output resistance soldered close to the electrodes, and four for the voltage differential drop measurements. This resulted in three possible voltage differential measurements. Each was soldered 5.5 cm apart and several tape widths from the electrodes to ensure a good current transfer distance [5]. Refer to Figure 1 for a graphical depiction of the voltage-tap layout.

3.4. HTS Tapes

The HTS tapes used were SuperPower Inc.'s SCS4050 (RE)BCO HTS tapes with dimensions of 4 mm x 0.1 mm [7]. The HTS tapes' transfer length measured from the edge of the anode to the cathode was 310 mm to within 0.5 mm accuracy. The tapes have a critical bending diameter of 11 mm and nominal critical current of 100 A at T = 77 K. Figure 3 shows the layer composition construction of the tape in order to achieve critical texturing of the (RE)BCO [7]. The soldering temperature withstand capability specified from the manufacturer is up to 250 ̊ C, however a slightly lower but sustained temperature may also unfavourably affect a tape's performance.

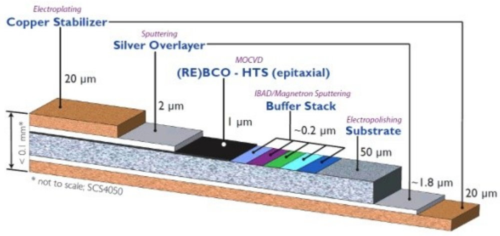

Fig. 3: Detailed diagram of layer composition of the SCS4050 HTS tape. Image is courtesy of Superpower Inc., *SuperPower® 2G HTS Wire Specification.*

3.5. HTS Tape Preparation

Three superconducting tapes, namely Tape A, Tape B and Tape C, were profiled at currents near and above their critical currents. Four voltage taps were soldered equidistant from the electrodes and 5.5 cm apart (see Figure 1). This allowed for three differential voltage tap measurements namely: Differential 1, Differential 2 and Differential 3. Tape A was exposed to 8 months of atmospheric conditions. The HTS tape is normally safely stored in a desiccator since prolonged atmospheric conditions can cause oxidation damage. Tape B was subjected to a small pinch or bend (stress-strain damage) at the centre of its length. The pinch-bend was imposed at about 90 degrees on 1 mm of its width with an approximate bend diameter of 1mm. Tape C was used as a control and not subject to any intentional damage.

4 EXPERIMENTAL RESULTS AND DISCUSSION

4.1. Destructive Thermal Quench

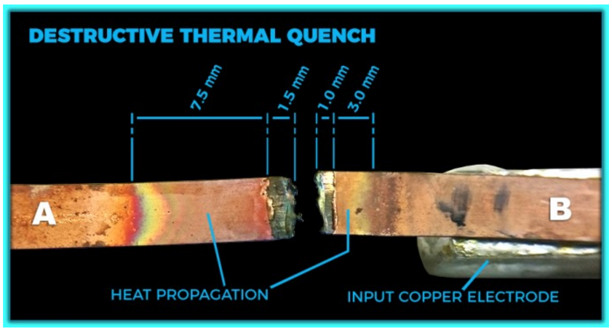

Fig. 6: Image of a destroyed tape after a destructive thermal quench.

Superconducting tape was subjected to current greater, 120 A, than the tape's critical current (= 112 A) in order to induce a thermal quench and long enough to be destructive, see Figure 6.

The thermal energy on Side B was partially absorbed by the copper electrode reducing the heat propagation distance as compared to Side A of the tape.

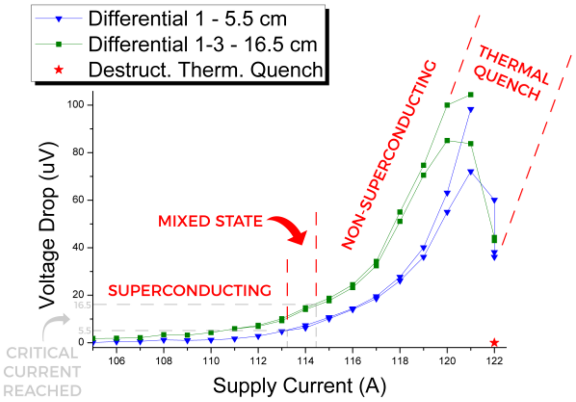

Fig. 7: Graph of voltage drop versus supply current for voltage tap 1 and tap 1-3 for the Destructive Thermal Quench Tape.

Measurements were taken for Differential Voltage 1 of 5.5 cm tap separation as well as Differential Voltage 1-3 of 16.5 cm tap separation for the thermal quench experiment. This means that the critical currents would be reached at differential voltages of 5.5 µV and 16.5 µV respectively, since the critical current occurs at 1 microvolt per centimetre [7] separation of the voltage taps. In Figure 7, the tape is superconducting up to a current of 113.2 A. Beyond this value it can be considered to be in a mixed state since not all of the 16.5 cm length monitored by the voltage taps has reach critical current. At 114.4 A the entire monitored length reaches critical current and enters a non-superconducting state. Once the current reaches 120 A, the voltage drop across the differential taps decreased suddenly. This behaviour is due to enough supercurrent flowing out of the superconductor material, (RE)BCO, and into the shunt copper material to heat it towards a thermal quench. Observation of this sudden decrease in differential voltage drop in the tap region could be integrated into a more sophisticated quench prevention circuit to predict possibly damaging thermal quench and shut off the power supply before the tape is destroyed; as opposed to just a detection circuit that shuts off power when there is a voltage detected above a threshold voltage. This would introduce specific tape behaviour monitoring which could suppliment traditional supply voltage threshold detection.

4.2. Differential Voltage Oscillations at High Currents

Throughout the experiments it was observed that as the current supply increased beyond the critical current, the differential voltage readings began to oscillate. In Figure 8 we can see that as the current increases far beyond the critical current of approximately 112 A, the voltage drop oscillates more drastically as the superconducting tape becomes unstable. This behaviour could be attributed to the *switching effect*.

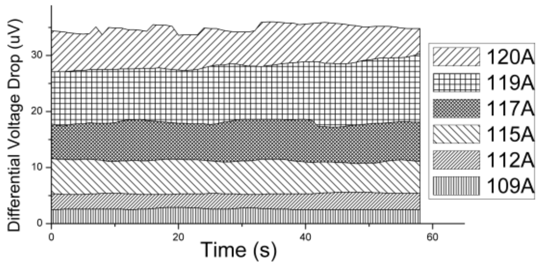

Fig. 8: Voltage versus time demonstrating oscillation effect.

To reduce the measurement error during experimentation with currents higher than the critical current, it is useful to measure the highest peak and the lowest peak and calculate an average.

4.3. Voltage-Current Damaged Tape Profiles

Each HTS tape, namely: Tape A (atmospheric oxidation damage), Tape B (bending damage) and Tape C (no damage), was injected with current below and beyond their respective critical currents. Three differential voltages were monitored for each tape, namely: Differential 1, Differential 2 and Differential 3. They represent three combination pairs of voltage taps, each 5.5 cm apart. This amounts to a critical current reached for a differential voltage of 5.5 µV.

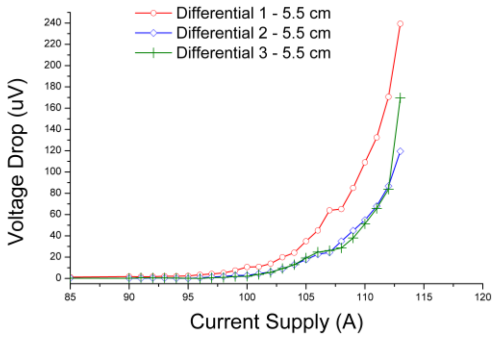

Fig. 9: Voltage versus current profile for Tape A.

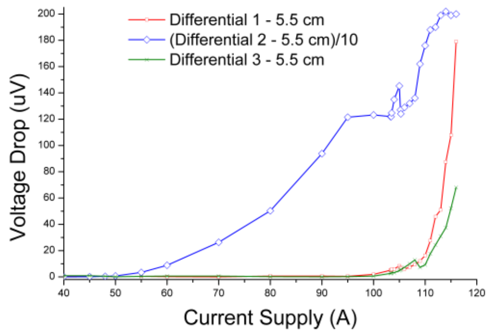

Fig. 10: Voltage versus current profile for Tape B.

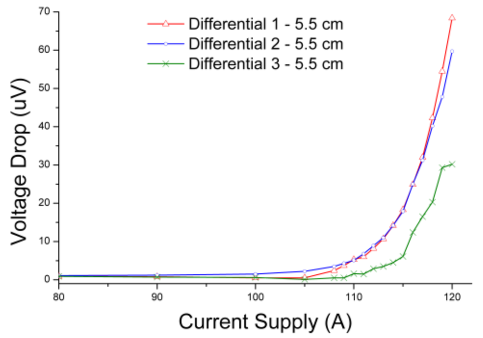

Fig. 11: Voltage versus current profile for Tape C.

Tape A's profile is shown in Figure 9. The critical current is observed to be below 100A, the rated critical current. The less steep graph, and premature critical current behaviour, indicates a damaged tape [5]. The damage is not excessive and the superconducting tape would still function correctly below 90 A (with a safety factor of below 0.9Ic).

Superconducting Tape B's profile can be observed in Figure 10. The Differential 2 region displays behaviour of a highly-damaged tape. The tape in this region had a critical current that was half that presented by Differential 1 and Differential 3. The bend damage on Tape B caused localised damage behaviour however the rest of the superconducting tape seemed to function sufficiently enough to assume relatively normal operation until higher than critical current values for these regions of the tape were reached. Excessive bubbling was observed at the bend region. Differential 3 displayed *switching effect* behaviour at higher than critical current values as a result of Differential 2 region's damage-induced and deviating voltage drop.

Tape C was not intentionally damaged in any way and its response can be observed in Figure 11. It is expected that all the differential voltage taps should behave similarly, as in the case of Differential 1 and 2, when a tape is undamaged, however Differential 3 showed divergent behaviour of a *switching effect* nature. This behaviour could be attributed to supercurrent flow in and out of the copper shunt outside of the voltage taps possibly due to heating damage incurred during contact soldering onto the anode. The affected region could only be situated in the region between the taps and the anode since the other regions' voltage-current curve exhibited a true curve whilst the Differential 3 region curve was shifted to a higher current as is synonymous with the *switching effect*.

4.4. Multiple Voltage Taps Discussion

The differential voltage measurement experiments all involved three differential voltage taps pairs being used to profile the superconducting tapes in question. This allowed for greater capability in indentifying the location, type and extent of any damage in the superconducting tape. During experimentation and development involving superconducting voltage taps, it is useful to have multiple differential voltage taps rather than just one as this allows greater investigative ability and problem definition resolution.

5 CONCLUSION

When considering the development of a High Temperature Superconducting (HTS) cable using HTS tapes, it is useful to have an understanding of the tapes' behaviour under extreme or damaging conditions.

A destructive thermal quench was investigated whereby it was found that the differential voltage of the voltage taps decreased at the advent of the thermal quench and continued to decrease right up until the superconducting tape was destroyed. The superconducting tape was found to exhibit increasing extremity of oscillatory behaviour above the critical current.

A superconducting tape that was exposed to atmospheric conditions for 8 months showed voltage-current profile behaviour indicative of a damaged tape. This was observed by a less steep differential voltage increase at a lower critical current. Another superconducting tape was partially bent at a point halfway along its length and in this region a critical current value was reached at approximately half that of non-damaged regions of the same tape. The non-damaged regions continued to operate normally but the region after the damaged region exhibited *switching effect* behaviour due to the damaged region's resistive behaviour. An increased heat generation was observed through excessive liquid nitrogen bubbling at the bend. A third tape was not damaged in any way but exhibited *switching effect* behaviour at the 3rd differential voltage region. This was presumed to be due to supercurrent flow in the copper shunt between the 3rd differential and the anode. It was posited that this may have been due to heating damaged incurred during anode contact soldering since the other regions behaved normally.

Throughout the experiments the differential voltage taps all exhibited variations in behaviour dependent on the condition of the superconductor tape in their respective regions. The implementation of multiple voltage taps can improve the ability to identify the location of the damage in a tape and the type and extent of the damage.

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7 REFERENCES

[1] R. L. Garwin and J. Matisoo, "Superconducting lines for the transmission of large amounts of electrical power over great distances," *Proc. IEEE*, vol. 55, pp. 538-548, 1967.

[2] P. M. Grant (2007, June). "Superconducting Lines for the Transmission of Large Amounts of Electrical Power Over Great Distances: Garwin–Matisoo Revisited Forty Years Later," *IEEE Transactions on Applied Superconductivity*, vol. 17, No. 2, pp. 1641-1647.

[3] Malozemoff, A., Maguire, J., Gamble, B., & Kalsi, S. (2002, Mar.). Power applications of high-temperature superconductors: status and perspectives. *IEEE Transactions on Applied Superconductivity*, vol. 12, No. 1, pp. 778–781. http://doi.org/10.1109/tasc.2002.1018517

[4] M. Noe, "Superconducting Cables," presented at the 13th European Conference on Applied Superconductivity, Geneva, Switzerland, 2017.

[5] J. W. Ekin, *Experimental Techniques for Low-Temperature Measurements*, vol. I. Oxford: Oxford University Press, 2006, pp. 3-4, 49-55, 368-372

[6] A. L. Mbaruku, U. P. Trociewitz, X. Wang, J. Schwartz (2007, June). "Relationship Between Conductor Damage, Quenching and Electromechanical Beaviour in YBCO Coated Conductors," *IEEE Transactions on Applied Superconductivity*, vol. 17, No. 2, pp. 3044-3049.

[7] Superpower Inc., Furukawa Electric Co., Ltd (2012), *Superpower® 2G HTS Wire Specifications*, 450 Duane Ave., Schenectady, NY 12304, USA. http://www.superpower-inc.com/system/files/SP_2G+Wire+Spec+Sheet_for+web_2012FEC_v2_1.pdf